

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-adad0b0c76a58d415.jsonl`  
- SHA-256: `982432dd3d26fc8e0101d601ff16ce09f6faccc0cd7524b78b43b44452e527fc`  
- Source modified: `2026-06-06T17:30:55+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:30:14.120Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** https://arxiv.org/abs/2306.04948

****Title:**** ShuttleSet: Human-Annotated Stroke-Level Singles Dataset for Badminton (KDD 2023)

****Found via:**** broad/primary search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:16.176Z)

I'll fetch the source page and extract relevant claims.

3. user (2026-06-06T17:30:27.013Z)

ShuttleSet: Badminton Dataset Summary

****Title:**** "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis"

****Authors:**** Wei-Yao Wang, Yung-Chang Huang, Tsi-Ui Ik, Wen-Chih Peng

****Publication Date:**** Submitted June 8, 2023; accepted to KDD 2023

Dataset Overview

****Size and Scope:****

The dataset encompasses "104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021" involving 27 elite men's and women's singles players.

****Shot Classification:****

The dataset includes "18 distinct classes" for shot types, along with "hitting locations, and the locations of both players at each stroke."

Annotation Methodology

The researchers employed "a computer-aided labeling tool to increase the labeling efficiency and effectiveness" during manual annotation at the stroke level for singles matches.

Supported Tasks

The paper provides "multiple benchmarks (i.e., stroke influence, stroke forecasting, and movement forecasting)" to demonstrate the dataset's utility for tactical analysis.

Access and License

The paper indicates a "CC BY 4.0" license, permitting broad use. The project page references GitHub (CoachAI-Projects) for potential access information.

4. assistant (2026-06-06T17:30:29.521Z)

I have the key details. Let me verify the license, as this is critical to the research question's commercial-licensing constraint, by checking the GitHub repository.

5. user (2026-06-06T17:30:36.853Z)

ShuttleSet Dataset License and Content

Based on the provided GitHub page, I can only determine limited information:

****License:**** The repository shows an "MIT license" link, suggesting the code uses MIT licensing. However, the page does not explicitly state whether a separate data license exists for ShuttleSet or what specific commercial use terms apply to the dataset itself.

****Dataset Content:**** According to the overview, ShuttleSet is described as "the largest badminton singles dataset with stroke-level records" published at KDD-23. The page indicates it contains stroke-level annotations but does not specify whether it includes rally counts, temporal segmentation timestamps, or rally start/end markers.

****What's Missing:**** The page provides no direct quotes about licensing terms, commercial restrictions, or detailed dataset specifications. For definitive answers about data licensing, commercial use permissions, and exact temporal annotation details, you would need to consult the actual ShuttleSet publication (referenced as available at arxiv.org/abs/2306.04948) or the dataset documentation within that repository folder.

6. assistant (2026-06-06T17:30:40.423Z)

I have enough to extract claims. The arXiv page (CC BY 4.0 license declared on arXiv, which is the standard arXiv submission license; the dataset itself is annotated via the CoachAI projects which use MIT for code). I'll ground the claims in what was directly confirmed.

7. user (2026-06-06T17:30:48.203Z)

Structured output provided successfully

8. assistant (2026-06-06T17:30:55.705Z)

Source extraction complete. Key findings on ShuttleSet (arXiv 2306.04948, KDD 2023, primary research source):

- ****Scale & content:**** 104 sets, 3,685 rallies, 36,492 strokes across 44 singles matches (2018-2021), 27 elite players; stroke-level labels with 18 shot-type classes plus hitting/player locations.
- ****Relevance to rally segmentation:**** Annotations are rally-organized (3,685 rallies), so rally structure exists ? but the dataset's benchmarks target tactical analysis (stroke influence, stroke/movement forecasting), NOT temporal rally start/end segmentation. It is not a turnkey segmentation dataset.
- ****Annotation effort:**** Built via manual stroke-level labeling with a computer-aided tool ? rich labels are not auto-derived from video.
- ****License caveat:**** arXiv paper is CC BY 4.0 and the CoachAI code repo is MIT, but the dataset's own data-use terms were not definitively stated on the pages fetched and should be confirmed at source before commercial reliance.

Tangential to the core "audio vs. pose vs. VLM-distillation" method comparison, but central as the candidate badminton dataset and relevant to the dataset-licensing sub-question.